

MAHMOUD MAHER EL-SAID

AI Student | Data Analysis | Machine Learning | Computer Vision | Reinforcement Learning

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Kaggle](#)

PROFESSIONAL SUMMARY

Artificial Intelligence student specializing in Intelligent Systems, with applied portfolio projects in Python, data analysis, machine learning, computer vision, reinforcement learning, and RAG systems. Comfortable working with datasets, notebooks, image-processing pipelines, model evaluation, and clear technical documentation. Seeking remote internships, junior AI/Data roles, and freelance opportunities.

TECHNICAL SKILLS

Programming & Tools: Python, Jupyter Notebook, Git, GitHub, VS Code, HTML/CSS/JavaScript basics

Data Analysis: Pandas, NumPy, data cleaning, EDA, visualization, Microsoft Excel, Matplotlib

Machine Learning: scikit-learn, supervised/unsupervised learning, model evaluation, feature engineering, classification, regression

Deep Learning & CV: TensorFlow/Keras, PyTorch basics, OpenCV, CNNs, image processing, segmentation, localization

RL & AI Apps: Q-Learning, SARSA, First-Visit Monte Carlo, Webots, Gymnasium, RAG, embeddings, Chroma, prompt design

EDUCATION

Arab Academy for Science, Technology and Maritime Transport | Egypt

Bachelor of Artificial Intelligence, Intelligent Systems Specialization | Expected Graduation: 2027

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, Data Science, Python Programming, Intelligent Systems, Data Analysis.

SELECTED PROJECTS

Financial Report RAG Assistant | *Python, Chroma, embeddings, Gemini, RAG*

- Built a retrieval-augmented generation workflow for financial report Q&A, including loading, chunking, embeddings, retrieval, prompt building, and response generation.
- Used Chroma and local embeddings to retrieve relevant document chunks and support source-aware answers; structured tests for core prompt/chunking logic.

Skin Lesion Segmentation and Localization | *Computer Vision, U-Net, TensorFlow/Keras, OpenCV*

- Developed a medical image analysis workflow for skin lesion segmentation using image/mask preparation, preprocessing, training, validation, and visual outputs.
- Organized predicted masks, before/after visualizations, and training artifacts to support clear project documentation.

Sudoku Solver from Camera | *OpenCV, image processing, digit recognition, backtracking*

- Created an end-to-end pipeline to detect a Sudoku board from an image, correct perspective, extract cells, recognize digits, solve the puzzle, and visualize the result.
- Applied thresholding, contour detection, perspective transform, grid extraction, and rule-based solving.

Reinforcement Learning GridWorld with Webots | *Q-Learning, SARSA, Monte Carlo, Webots*

- Implemented and compared Q-Learning, SARSA, and First-Visit Monte Carlo in a grid-world with obstacles, rewards, episodes, and policy tracking.
- Integrated learned behavior into a Webots simulation and logged actions, rewards, and path progress to explain agent decisions.

SimpleML Library from Scratch | *NumPy, ML fundamentals, reusable code*

- Built a small educational ML library from scratch to practice algorithm implementation, training logic, and reusable component design.

CIFAR-10 Image Classifier API | *CNN/ResNet, FastAPI, Python*

- Prepared an image classification workflow and a simple FastAPI inference interface, practicing preprocessing, evaluation, and deployment-oriented structure.

CERTIFICATIONS AND TRAINING

AMIT Beyond Education - Master Machine Learning & AI Diploma: 220 hours; Grade 92%; topics: Python, Data Science, ML, Deep Learning, Computer Vision, NLP; Jan 2025 - Aug 2025.

Coursera / MBRF / UNDP - Machine Learning - L1 Badge: Verified learner badge completed Aug 2025; covered supervised learning, unsupervised learning, neural networks, model evaluation, TensorFlow, scikit-learn, and RL fundamentals.

LANGUAGES

Arabic: Native | **English:** Professional working proficiency / improving through technical study and freelance communication